

Abdullah Mahmoud Fawzy

Frontend / Fullstack Engineer

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PROFESSIONAL SUMMARY

Software Engineer with a Bachelor's degree in Mathematics from Helwan University and a strong analytical foundation in problem solving, logic, and algorithmic thinking. Specializing in modern frontend and fullstack engineering with **Next.js, TypeScript, React, and Node.js**. Experienced in designing maintainable web architectures using **SOLID principles and Strategy/Factory patterns**, building real-time systems, implementing secure authentication and request protection, and optimizing server/client data flows.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend: Next.js (App Router, Server & Client Components, Server Actions), React, React Hooks, Redux Toolkit, TanStack Query

UI & Styling: Tailwind CSS, CSS Modules, Responsive Design, Accessible UI (ARIA), Recharts

Backend & APIs: Node.js, Express.js, RESTful APIs, WebSockets (Socket.IO), JWT Authentication, Zod Schema Validation, Rate Limiting

Databases & Cloud: PostgreSQL, Supabase, MongoDB, Mongoose, Vercel

Architecture & Engineering: SOLID Principles, Strategy Pattern, Factory Pattern, Optimistic UI Updates, Server-Side Prefetching & Client-Side Hydration, Git & GitHub

FEATURED ENGINEERING PROJECTS

1. Devfolio AI — Engineering Showcase & Architectural Assistant

Next.js 15, TypeScript, Tailwind CSS, Strategy Pattern, Rate Limiting

- Architected an engineering-focused portfolio platform with dynamic case studies and an interactive AI technical assistant.
- Decoupled LLM provider integration using **Strategy and Factory patterns**, enabling provider-agnostic integration and simplifying the addition or replacement of providers.
- Applied **Single Responsibility and Open/Closed principles** to separate provider-specific logic from the application layer.
- Implemented an in-memory sliding-window **rate limiter** for serverless API routes to mitigate abusive traffic, excessive requests, and downstream API quota exhaustion.
- Enforced end-to-end type safety with strict TypeScript configuration and schema-based request validation.

2. Inventory Management Dashboard

Next.js 16, TypeScript, TanStack Query, Tailwind CSS, Recharts

- Engineered an inventory and analytics dashboard using Next.js App Router and TanStack Query.
- Implemented **server-side data prefetching with client-side cache hydration**, improving initial data availability and avoiding redundant client-side requests.
- Built multi-parameter filtering, sorting, debounced search, and client-side pagination across **1,000+ inventory records**.
- Developed interactive analytics modules for revenue trends, low-stock alerts, and category distribution using Recharts.

3. Real-Time Team Chat & Collaboration Application

React, Redux Toolkit, Node.js, Express, MongoDB, Socket.IO, JWT

- Built a fullstack real-time messaging application with persistent conversations and active user presence.
- Engineered bidirectional communication using **Socket.IO and room-based broadcasting**, with optimistic client-side updates for responsive messaging.
- Implemented stateless **JWT authentication** using HTTP-only secure cookies, bcrypt password hashing, and protected Express middleware.
- Managed complex client-side conversation state with **Redux Toolkit**, including active conversations and unread message notifications.

4. Hotel Management & Operations Dashboard

React, Supabase, PostgreSQL, React Query, React Hook Form, Recharts

- Developed an internal operations dashboard for managing bookings, cabin inventory, and guest check-in/check-out workflows.
- Integrated **Supabase PostgreSQL** with Row Level Security (RLS) policies for controlled data access.
- Implemented structured form validation with **React Hook Form** to prevent invalid reservation submissions and provide clear client-side error feedback.
- Built administrative analytics covering occupancy, revenue, and stay-duration metrics.

EDUCATION

Bachelor of Science in Mathematics

Helwan University, Cairo, Egypt — Graduated October 2024

LANGUAGES

Arabic: Native

English: Professional Working Proficiency